

Stock analysis you can actually verify.

Product & Technical Documentation

For Indian retail investors · v1.0 · September 2026

In this document

- What Veridex does and why it matters
- How users can trust the AI (grounding, methodology visibility)
- Comparison with Groww, Zerodha Kite, INDmoney, TickerTape, Screener.in
- System architecture, tech stack, data flow diagrams
- Security model, compliance stance, and non-goals

Table of contents

1. Executive summary
2. The problem — why Veridex is needed
3. What Veridex does
4. Trust — how you know the AI isn't making it up
5. Competitive analysis
6. System architecture
7. Tech stack
8. Feature deep-dive
9. Grounded AI chat — end-to-end data flow
10. Security & data model
11. Compliance & non-goals
12. Sources & references

1. Executive summary

Veridex is a dashboard for Indian retail investors that treats verifiability as a first-class feature. Every score, signal, and AI-generated sentence traces back to a number the backend just computed or freshly fetched from a licensed source. Nothing is predicted with hidden math. Nothing is invented from a language model's training data.

The platform is intentionally not a broker. It never places buy or sell orders on your behalf. It gives you live prices from Upstox, a transparent fundamentals screener, sentiment and volume-anomaly signals with the methodology visible next to every number, and an AI chat that refuses to answer questions with numbers it wasn't given.

Live at <https://veridex-lake.vercel.app>. Source at github.com/sahilkarande0918-cmd/Veridex.

2. The problem — why Veridex is needed

India has ~180M demat accounts. Most retail investors get their calls from three sources: Telegram tip channels, YouTube 'gurus', and broker apps that surface trending stocks. None of these show their working. When a source says *"buy this, target 20% in 3 months"*, there is no way for a retail investor to audit whether that pick has any track record, whether the model was tested on data it had never seen, or whether the reasoning came from an LLM that hallucinated a P/E ratio.

The gaps in the existing landscape:

- **Screener.in** exposes raw fundamentals brilliantly, but has no live prices, no signals, and no explanation layer for beginners.
- **TickerTape** gives good scorecards but its methodology (weights, universes, cut-offs) is proprietary — you can see the number, not the maths.
- **Groww / Zerodha Kite / Upstox** are broker apps: their goal is order flow, not education. Their 'insights' widgets are opaque and often marketing-driven.
- **INDmoney** is excellent for aggregation and net-worth tracking, but analysis is thin and non-configurable.
- **Every AI chatbot bolted onto finance apps** hallucinates numbers because the model is asked to answer directly instead of being handed live data first.

Veridex fills the 'show your work + refuse to lie' gap between these categories.

3. What Veridex does

The dashboard has eight sections. Each one is opinionated about verifiability.

- **Overview** — live KPI strip (portfolio value, P&L, Nifty 50, top mover), plus movers, latest news, and cards for Signals and AI Chat. Every card is a link into its underlying page.
- **Profile** — INDmoney-style net worth summary: total net worth (stocks current + cash), invested, unrealised P&L, positions count, sector allocation donut, top holdings table with live LTP, best and worst performer, and stored preferences (risk / horizon).
- **Charts** — TradingView Lightweight Charts candlestick view over Upstox data, 30-minute / 1D / 1W / 1M intervals, and a 5-second live LTP poll on the selected symbol.
- **Portfolio** — manual holdings entry (symbol, qty, buy price, buy date), live P&L per row, ApexCharts allocation donut. Row-level security scopes every read/write to your own account.
- **Screener** — 30 Nifty heavyweights ranked by a 40% value / 20% leverage / 40% growth composite. Every subscore, every underlying number (P/E, P/B, D/E, revenue growth), and the medians used to normalise them are visible on the same row as the score. Capital-aware filter hides stocks outside your budget.
- **Signals** — 20-day rolling z-score anomaly detection on price and volume (flags at $|z| \geq 2$), plus a Groq-classified headline-tone panel that shows the per-headline label AND the rationale AND the source link.
- **News** — RSS aggregation from Economic Times, Moneycontrol and LiveMint, refreshed every five minutes. Company / ticker search hits NewsData.io. An IPO filter surfaces IPO-tagged items.
- **AI Chat** — a grounded assistant. Given a user message, the backend runs a keyword-rule intent router, fires the relevant live data fetches in parallel, and only then hands the results to Groq with a strict system prompt: *never invent a market number*.

4. Trust — how you know the AI isn't making it up

Four mechanisms enforce this, layered.

4.1. Grounded prompting, always live-fetch first. The chat Edge Function never sends a user message straight to Groq. It first routes the message (keyword rules, no LLM), then fetches the data that intent needs — Upstox for market questions, NewsData for news, your own holdings (via your JWT, so RLS enforces access) for portfolio questions. That live JSON is embedded in the system prompt as a CONTEXT block. The model is instructed that any number it emits **MUST** come from that block. If it isn't there, the correct answer is "I don't have that right now — I can pull it if you'd like."

4.2. Every answer shows what it was grounded on. The response includes a `grounded_on` object — the UI renders it as green pills under each assistant message: *"grounded: live market"*, *"grounded: your holdings"*, etc. If a question didn't trigger a live fetch, no pill appears — a visible signal that the answer is generic, not backed by fresh data.

4.3. Every score has its formula next to it. The Screener's "Show methodology" toggle expands the exact mapping used: "lower is better — value/median → 100 when far below median, 50 at median, 0 at 2× median", plus the current cross-section medians (P/E, P/B, D/E) used as reference. Signals cards state the window (20 days), the metric (z-score of daily returns / volume), and the threshold for flagging.

4.4. No look-ahead bias in any time-series work. The anomaly detector excludes today's bar from the reference window it standardises against, so a spike today can never be self-normalised into looking normal. There is no forward-looking "prediction" model in the current build — signals are strictly descriptive of recent behaviour.

4.5. Transparent 'sample' labelling on the screener. The fundamentals file that seeds the screener is illustrative round-numbers for a demo; the UI carries a persistent amber banner saying so and naming the three live-feed swap paths (Alpha Vantage OVERVIEW, FMP, IndMoney fundamentals). The scoring engine and UI are production-ready — only the data source is a stub, and the stub is called out.

5. Competitive analysis

A brief scan of what each incumbent optimises for, and where Veridex sits in the gap.

Platform	Primary strength	Weakness Veridex addresses
Screener.in	Deep 10-yr fundamentals, custom queries	Noise, price, no explanation for beginners, no AI layer
TickerTape	Proprietary Scorecard + Market Mood	Scorecard methodology is opaque; you see numbers, not the maths
Groww	Broker + MF onboarding, huge user base	Insights are marketing widgets; goal is order flow, not audit
Zerodha Kite	Fastest execution terminal in India	Pure trading UI, no research layer, no fundamentals context
INDmoney	Net-worth aggregation across accounts	Analysis thin, not configurable, no methodology disclosure
Moneycontrol	News + basic tracking	News-first, no verifiable analytics, ad-driven
Telegram / YouTube tips	Free, entertaining	Zero accountability; no backtest, no track record, no methodology
Veridex	Show-your-work analysis + grounded	(This is the delta) — every number cites its formula and source

Veridex is not trying to replace any of these — it is deliberately not a broker, not a MF platform, and not a news publisher. It is the audit-and-explain layer that sits alongside them.

6. System architecture

Vercel serves a static React bundle. Every side-effectful or credential-holding call is a Supabase Edge Function (Deno). Supabase Postgres holds user-scoped rows with row-level security. Third-party APIs (Upstox, Groq, NewsData, RSS, FCM) are never called from the browser — the frontend only calls Edge Function URLs.

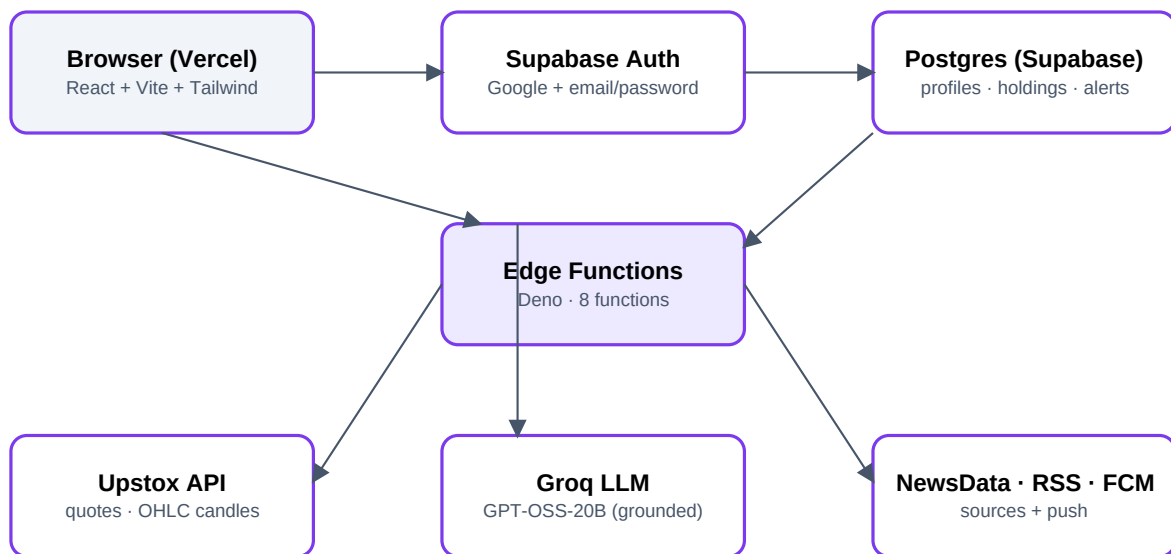


Fig 1 — Veridex system architecture. The browser never sees a credential; every third-party call is proxied through a Supabase Edge Function.

7. Tech stack

Every choice below optimises for a small operating budget, a small surface area of secrets, and no vendor lock-in on the analytical layer.

Frontend

React 18 + Vite 5	SPA. Fast HMR in dev, code-split chunks in prod.
TypeScript 5.6 (strict)	Static typing across UI + API client.
Tailwind CSS v4	Utility-first styling. Data-theme swap for light/dark.
React Router 7	Declarative routing, nested layouts, lazy routes.
Framer Motion 13	Micro-motion on the dashboard (stagger, spring-hover, count-up).
Lightweight Charts 5	TradingView candlestick charts.
ApexCharts 7	Portfolio allocation donut.
Firebase Web SDK 12	FCM push registration + foreground message handling.
@supabase/supabase-js 2	Auth session + RLS-protected reads/writes.

Backend (Supabase)

Supabase Postgres 15	Tables: profiles, holdings, watchlist, alerts. RLS on every table.
Supabase Auth	Email/password + Google OAuth.
Supabase Edge Functions	Deno runtime. Eight functions (see §8).
pg_cron + pg_net	Daily 06:00 UTC schedule for the review-nudger.

Third-party services

Upstox v2 API	Real-time LTP, batched quotes, historical candles. Read-only token.
Groq	openai/gpt-oss-20b (chat + JSON sentiment). Low latency, low cost.
NewsData.io	Company-specific news search over Indian business feeds.
Public RSS	Economic Times / Moneycontrol / LiveMint aggregated latest feed.
Firebase FCM	Web push. HTTP v1 API with service-account JWT signing in Deno.

Hosting & CI

Vercel (Hobby)	Static hosting. Auto-deploys every push to main.
Supabase (Free tier)	Postgres + Auth + Edge Functions in one project.
GitHub	Single main branch. Free tier.

8. Feature deep-dive

Every server-side action lives in exactly one Edge Function.

Function	Method	What it does
upstox-quote	GET ?key=...	Live LTP for one instrument. Polled every 5s by the Charts page.
upstox-candles	GET ?key=...&interval=...&days=...	Historical OHLC. Returns ascending array for TradingView.
market-summary	GET	One batched Upstox call → Nifty + 20-stock universe + gainers/losers. 30s
news-feed	GET	Parallel fetch of three RSS sources. Regex parser, no XML dep. Per-source
news-search	GET ?q=...	NewsData.io company query (India, business, English).
sentiment	GET ?q=...	News → Groq structured JSON labeling → aggregate counts + per-headlin
chat	POST	Intent router → parallel live fetches → grounded Groq call.
nudge-reviews	POST (cron)	Finds holdings older than 30d, idempotent insert of review alerts, FCM push

9. Grounded AI chat — end-to-end data flow

The chat pipeline is deliberately deterministic where possible and leaves only the natural-language shaping to the model.

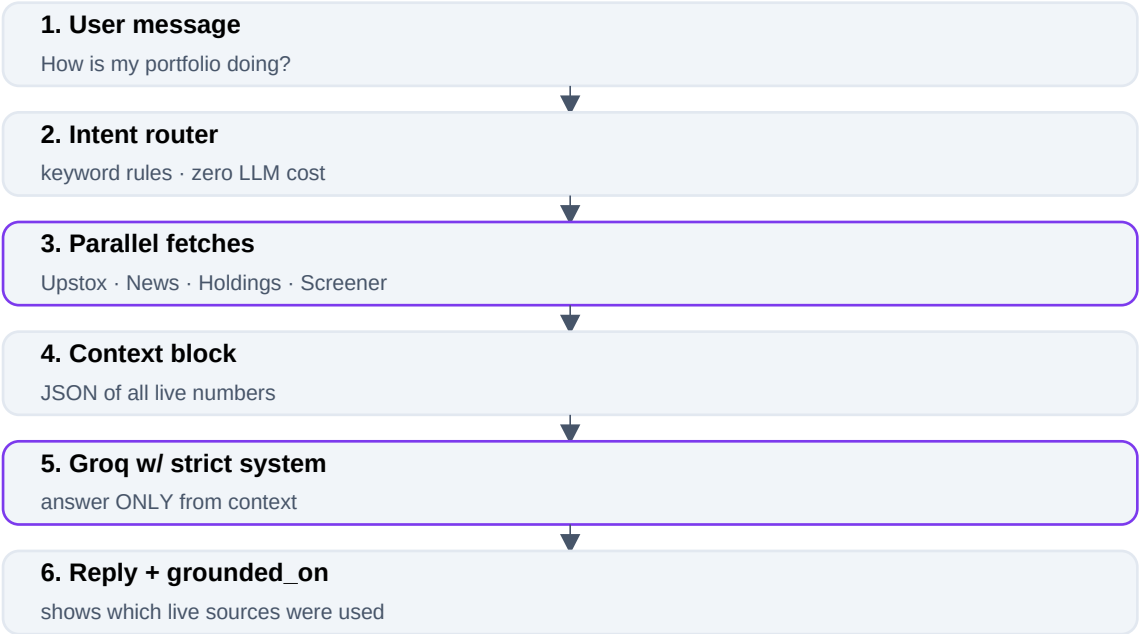


Fig 2 — Every AI reply follows this six-step pipeline. Steps 3 and 5 (the violet-bordered ones) are where all trust decisions are made.

What this prevents:

- Numeric hallucination — the model never sees the user’s message alone, only alongside the CONTEXT block.
- Stale data — every eligible answer triggers a live fetch, so recall from training is never the answer to a market question.
- Silent recommendations — ‘what should I buy?’ routes to the Screener top-3 pointer with methodology, not to an LLM opinion.
- Cross-user data leakage — portfolio reads use the user’s own JWT via the Supabase REST API, so RLS enforces scope.

10. Security & data model

Values are partitioned by where they live and who can read them.

Server-only (Supabase Edge Function secrets)	SUPABASE_SERVICE_ROLE_KEY, UPSTOX_ACCESS_TOKEN, GROQ_API_KEY, NE
Client bundle (safe by design)	VITE_SUPABASE_URL, VITE_SUPABASE_ANON_KEY (RLS enforces access), Firebas
Local only, gitignored	.env, veridex-*firebase-adminsdk*.json

Row-level security is on for every table. Policies restrict reads and writes on profiles, holdings, watchlist and alerts to rows where `auth.uid()` matches `id` or `user_id`. Even the service-role key is used only by the

nudger Edge Function, never returned to the client.

11. Compliance & non-goals

Not SEBI-registered investment advice. Veridex is positioned as an educational and analytical tool. This disclaimer is persistent on every screen that shows a signal, a score, or an AI-generated summary. The AI system prompt requires the model to end every substantive answer with the same disclaimer.

Non-goals (explicit, intentional):

- No buy/sell order placement, ever. The Upstox integration uses a read-only token and does not exercise the trading endpoints.
- No personalised recommendations from the LLM. Recommendations, when asked, route to the Screener's transparent composite score.
- No forward-looking price predictions. Every quantitative signal is descriptive of recent behaviour.
- No aggregation of user activity for third parties. Analytics are limited to what Supabase and Vercel collect by default.

12. Sources & references

Competitor scan (September 2026 web sources):

- Winvesta — *Fundamental analysis tools and screeners: 2026 guide for India* · winvesta.in/blog/investors/fundamental-analysis-tools-and-screeners-2026-guide
- Sharpely — *Best Stock Screener in India 2026* · sharpely.in/blogs/best-stock-screener-india-2026
- Sharpely — *Compare India's Top Investment Research Platforms* · sharpely.in/compare-platforms
- Uninvest — *Best Stock Analysis App India 2026* · uninvest.in/blogs/best-stock-analysis-app-india-2026-top-7-picks-for-research-screeners-amp-advisory
- Top Stock Market Broker — *TickerTape Review 2026* · topstockmarketbroker.com/2026/08/tickertape-review-2026-app-features.html

Vendor documentation and specs:

- Upstox API v2 — *Market Quote and Historical Candle* · upstox.com/developer/api-documentation
- Groq — *OpenAI-compatible chat completions* · console.groq.com/docs
- Supabase — *Edge Functions, RLS, pg_cron* · supabase.com/docs
- Firebase Cloud Messaging — *HTTP v1 API* · firebase.google.com/docs/cloud-messaging
- NewsData.io — *Latest news search endpoint* · newsdata.io/documentation